# SHOULD I BE AN ENGINEER?

Everything You Need to Choose Your Engineering Path




James R. Martin, Jr., PhD, PE

*“For all good teachers and good students—they have both taught me more than I knew possible.”*

**James Martin**

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# FOREWORD

## Why I Wrote This Book

When I was a kid and needed help with math, I would ask my mom first, sometimes I would ask my dad. They were both great with numbers, but they were accountants, so when I hit algebra, they had to call in the family's only engineer, Uncle Jay.

Jay and Dad were early computer adopters. They were writing code to create spreadsheets long before Excel or even Lotus were around. I was there, at age five, learning along with them. It was cool to be around the cutting edge of computers, surrounded by all their code and sketches and stuff. In my mind, Uncle Jay was the smartest person on Earth.

When I was thirteen, I was blessed to be admitted into an advanced, highly competitive, academic curriculum. I had no idea what was waiting for me. I was a good student throughout grammar school. However, that all changed when I got to high school. The first four months were an exercise in relearning how to learn and how to study.

Enter Uncle Jay, the family engineer and math tutor. After talking with him for about an hour, I went from being clueless about algebra to being up to speed. After that, I didn't make anything short of a high A in math, be it algebra or differential equations. It's clear to me today that this only happened because my uncle was able to translate the world of mathematics into a language that a teenager could grasp. That's when I started thinking "How cool it must be to be an engineer, it must be like a superpower."

As I moved through school, teachers would regularly tell me that my talents in science and math would make me an excellent engineer. Combine that with Uncle Jay being the smartest guy I knew and my desire to be just like him, and my career path was set.

Yet, despite my certainty, I never thought of asking the most important question: what do engineers do every day? I never knew to consider the broad array of subdisciplines within the engineering profession. I simply pointed myself toward my engineering studies and didn't stop until I had three different engineering degrees—all without ever looking up to know if I would like the work that I would be doing. I was good at learning, and I found fulfillment in mastering the complexities of fluid mechanics, but I didn't know what I would be doing the day I walked off campus.

I wrote this book so you can ask yourself the questions I never considered, so that you can identify the engineering subdiscipline that fits you before you commit years to pursuing an expensive degree. Between these pages you will find all the information you need to decide which engineering subdiscipline is best suited to you and your interests.

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# Is Engineering for You?



**Are you naturally curious?** Do you find yourself taking things apart just to see how they fit together, asking, "How does this work?" Do you enjoy puzzles, math, or science? Not just for grades, but because solving problems feels rewarding. If so, you are already thinking like an engineer.

A good engineer is someone who possesses a unique blend of persistence, creativity, and a desire to improve the world around them. They are the people who look at a broken system and see an opportunity, and who understand that the best solutions require both rigorous logic and out-of-the-box thinking.

Engineering isn't for everyone, and that's a good thing. The world needs poets, teachers, and historians, as much as it needs bridge builders and software designers. But if you're still holding, reading, or listening to this book, then you're wondering if you belong in the group of people that proudly designs, builds, and improves the systems upon which all of us depend. My goal is to help you answer that question using the experience of people that have already been down these roads and chosen which forks to take.

Engineering is built upon the knowledge of our predecessors, so why shouldn't your decisions be based upon the same. Isaac Newton wrote

> *"If I have seen further than others, it is by standing upon the shoulders of giants."*

Let's stand on some shoulders and see what we can learn.

# What Engineering Is—And What It Isn't

At its heart, engineering is disciplined problem-solving—applying math, science, logic, and judgment to real-world challenges. Done well, it quietly shapes daily life: the water we drink, the roads we drive, the apps on our phones. Engineering, unlike pure science, is not abstract theory. It's applied.

Many people view engineering as mathematical creativity confined by budgets, physics, safety margins, and deadline constraints. For many, this is what makes it fun—taking challenges and turning them into tangible solutions. If you enjoy solving puzzles with boundaries, engineering might just fit you perfectly. It's a career where you can rise and become a respected authority in the modern world.

Engineering isn't invention for invention's sake. Engineers don't dream up solutions and hope a problem comes along. It isn't art, though good design can be elegant. It isn't guesswork, intuition, or inspired brilliance—it's method, iteration, and accountability. Most engineering happens quietly, in the margins of spreadsheets and the fine print of specifications, long before anything gets built. The drama, when it comes, means something went wrong.

Make no mistake, it takes time to climb the ranks. New engineers must prove themselves through consistency and humility. But the reward is deep. It's the satisfaction of knowing your design holds under pressure, the adrenaline of clearing a project for an emergency response and the quiet pride of pointing to a structure or a system and telling your kids, "That only existed on my desk five years ago, and now the world uses it every day." It's in these moments that you're more than a worker, you become a contributor to society with a deep sense of fulfilment.



# The Engineering Families

**Choosing an engineering path** is a bit like exploring a new city—having a map helps. In this section, we'll take a deep dive into the five major families of engineering: Civil, Mechanical, Electrical, Chemical, and Software & Computer, along with their key subdisciplines. Each of these major families has its own "flavor," from designing the structures and machines that shape our world to developing the software and electronics we depend on every day.

This section provides a big picture overview of what each engineering family encompasses—why it matters, what it entails, and where you'll see it in action. As you read, take note of the role descriptions that resonate with you most. You might be drawn to designing huge infrastructure, harnessing electricity, creating new materials, or writing code; look for what interests you most, whatever that might be. The goal is to give you a genuine feel for what each subdiscipline and its specialties entail so you can choose a path that suits you.

There is no one-size-fits-all model for categorizing engineering roles, and each subdiscipline within each family could be a book of its own. For instance, a structural engineer (which is already a subdiscipline of the civil engineering family) could be a bridge engineer that only works on steel bridges, or a specialist that works on historic buildings. They are both structural engineers, but their clients have zero overlap, their workflow has nearly zero overlap, and the pace of their work has nearly zero overlap.

Before we get into what makes each engineering family unique, it helps to recognize what they all have in common. No matter which family or subdiscipline you choose you'll run into a few shared realities.

- **Journeys all begin with an ABET Undergraduate Degree.**
- **All engineering roles are sought after and offer salaries above the national average.**
- **Every subdiscipline rewards attention to detail.**
- **Career advancement is universally possible through technical expertise as well as management.**



- **Select roles work alone but most require collaboration with multidisciplinary teams.**

You'll see these disciplines overlap in the case studies on page 142.

**A NOTE ON THE SALARY SCALES** The row of dollar signs on each profile shows where that discipline's typical pay sits relative to the other disciplines in this book—the low end of the scale is not minimum wage. Every engineering discipline here pays well above the US median wage and the pay in every one of them grows substantially with experience. Use the scales to compare disciplines with each other, not against the job market at large.

**Now, let's dive into the engineering landscape and see what each family is all about.**



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# STRUCTURAL ENGINEERING

*"If it rises from the ground and stands, a structural engineer made sure it could."*

**Skyscrapers, stadium roofs, and long-span bridges** don't stay standing by chance—they're the work of structural engineers who design the frameworks that keep them safe and stable. These engineers are tasked with ensuring that buildings, bridges, towers, and other structures can support the loads and stresses of everyday use, as well as extreme events, remaining functional, durable, and secure for decades to come.

## Fast Facts

**Education:** Electives in finite element analysis, foundations, and wood design; a master's degree is beneficial for advancement.

**Key Skills:** Strong foundation in math and physics, spatial visualization, precision, attention to detail, proficiency with design software, clear communication, and teamwork.

**Typical Employers:** Engineering consultancies (often small, specialized firms), construction companies, government agencies, integrated architecture and engineering firms.

**Personality Fit:** Detail-oriented, conscientious about rules and codes, collaborative team player, and calm under pressure.

**Salary Range:**

$ $ $ $ $ $ $ $ $ $

**Work Environment:**

Office Field

## Job Options

**Building Design Engineer:** Plans and designs the "skeleton" of buildings, ensuring foundations, beams, and columns can safely support all their loads.

**Bridge Engineer:** Designs bridges of all sizes, from small road overpasses to long-span suspension bridges, ensuring they can safely carry traffic and resist wind and weather.

**Marine/Offshore Engineer:** Designs structures in, on, or near water, like piers, ports, and offshore platforms, that can withstand waves, currents, and corrosive marine conditions.

**Special Structures Engineer:** Tackles unique projects, such as stadiums, large towers, and roller coasters, applying structural principles to ensure that the structures stay safe.



## Skillset Requirements

A structural engineer needs a mix of technical mastery and practical intuition. They must be adept at math-based analysis, specifically in calculating forces and load paths. Proficiency with computer-aided design (CAD) and analysis software is essential for modeling and testing structural behavior in digital environments.

Attention to detail is vital, since a small mistake in a load path assumption can overstress a critical connection. Strong communication skills are also important for collaborating with architects, contractors, and other engineers. Familiarity with building codes and standards, as every design must meet safety regulations. Through years of experience, these codes become second nature.



## Career Trajectory

Structural engineers begin their careers supporting calculations and assisting in drawings for modest projects. As they gain experience and licensure they take more responsibility, managing the design of entire projects and signing off on safety-critical plans. With time, many begin to specialize in earthquake or wind engineering, forensic work that investigates why structures failed, or retrofitting historic buildings. Others move into firm leadership or public sector roles, where they set standards and review designs for major infrastructure.



## Pace of Innovation

Innovation in structural engineering tends to be slow, steady, and safety driven. Today, advancements in computer simulations and Building Information Modeling (BIM) allow engineers to test and refine designs virtually before building. Future engineers will also start exploring new sustainable materials as new, eco-friendly alternatives to steel and concrete, such as new forms of engineered timber, emerge. Progress may seem slow from the outside, but innovation is continuous. Structural engineers can expect modern methods to evolve continuously as new techniques and materials hit the market each year.



# The Life of a Structural Engineer

**The Work:** Structural engineers ensure that all kinds of structures stay safe and standing. They turn design ideas into frameworks capable of supporting weight and resisting forces like gravity, wind, and earthquakes.

## Applications

Structural engineers calculate how much weight a structure needs to support, and choose appropriate beams, columns, or other components to ensure it can do so safely. Using computer simulations, they test how a design will perform under challenging conditions and adjust the plans until they meet all regulations. Once approved, they produce detailed drawings for builders that ensure all building code requirements are met. During construction, structural engineers continue to visit the site to inspect critical elements, solve unexpected problems, and answer questions, making sure the structure is built correctly and safely.

## Knowledge Base

These engineers combine principles of physics and advanced math to calculate forces and stresses. They study how different construction materials (such as steel, concrete, and wood) behave under loads, which helps them choose the right materials for each project. Structural engineers also learn structural analysis methods and specialized software in order to model and predict how designs will perform. Equally important, they learn building codes and standards so that every design meets safety regulations.

## Interdisciplinary Overlap

Structural engineers work closely with many other professionals. They collaborate with architects to make sure buildings match their design vision. They coordinate with mechanical, electrical, and plumbing engineers so that pipes, ducts, and cables can fit without compromising structural elements. Geotechnical engineers are consulted to understand soil conditions and design proper foundations for the structure. During the construction phase, structural engineers often interact with construction managers and contractors, addressing on-site issues or answering questions to clarify the plans. Structural engineers remain in constant communication with the whole project team from start to finish.

## Work Environment & Lifestyle

Most structural engineers work primarily in an office setting, using computers to run calculations, create models, and draft plans, with most meetings with teammates and clients conducted online or in meeting rooms. Once projects start, they're expected to visit construction sites to inspect progress and ensure the work matches their designs. The job usually follows normal weekday hours, but as deadlines approach, or if unexpected issues arise on site, extra hours might be required.

## A Day in the Life

### Meet Alex, a structural engineer

Alex is a structural engineer at a design firm, and no two days are the same. His work is a constant balance between the precision of the office and the unpredictable reality of the construction site.

**9:00 AM:** Alex's day begins with an urgent issue arriving from a job site: some anchor bolts for a building's steel columns are misaligned. He quickly reviews the plans and emails back a fix so the crew can proceed without delay. When he's done, he moves over to work on calculations for a community center roof, using design software to ensure the beams can handle heavy snow loads.

**12:00 PM:** At midday, Alex meets with an architect and a mechanical engineer to discuss a design change on his current project. The architect wants a wide, open lobby without interior columns. They brainstorm solutions, and Alex suggests using a steel truss above the ceiling to span the lobby. The architect agrees, since it keeps the space open, yet structurally sound—exactly what his design calls for.

**2:30 PM:** Grabbing his hard hat, Alex heads to a library project for a site visit before a concrete floor slab is poured. He walks the site, inspecting the steel rebar grid in the formwork and compares it to the drawings. Once he is satisfied that the reinforcement is correctly placed, he gives the go-ahead for the concrete pour.

**4:00 PM:** Back at the office, Alex writes a brief report on the site inspection and then spends the final hours of the day reviewing a junior engineer's calculations. After a few minor tweaks, he signs off that everything looks good.

Alex leaves knowing that structural engineering is equal parts analysis, judgment, and responsibility. His work may not always be visible once a building is complete, but every decision contributes to structures that are safe, durable, and built to last for generations.

# ROBOTICS ENGINEERING

*"If it thinks and moves on its own, a robotics engineer coded its brain and built its frame."*

**Robots once belonged to science fiction;** today they weld cars, assist surgeons, explore other planets, and even vacuum the floor at home. Behind each of these machines stands a robotics engineer. As robots take on more of the work that is dangerous, repetitive, or beyond human reach, the engineers who design them are in growing demand. The work is a rare blend of code and hardware: robotics engineers write the software that lets machines sense their surroundings, decide, and act—doing real work on their own.

## Fast Facts

**Education:** Electives in robotics, kinematics, control systems, mechatronics, and computer vision; master's degree helpful and preferred for research-heavy roles.

**Key Skills:** Programming in Python and C++, proficiency with ROS and simulation tools, a strong grasp of kinematics and control theory, machine learning and computer vision, plus hands-on hardware debugging and teamwork.

**Typical Employers:** Industrial automation and robotics firms, automotive and warehouse logistics companies, medical device and aerospace makers, defense contractors, and big-tech research labs.

**Personality Fit:** Curious tinkerer, comfortable bridging hardware and software, a patient debugger, and calm under pressure.

**Salary Range:**

$ $ $ $ $ $ $ $ $ $

**Work Environment:**



Office Lab Factory

## Job Options

**Robotics Software Engineer:** Writes and tests the code that controls a robot's perception, planning, and motion, usually in Python or C++ using ROS.

**Automation Engineer:** Programs and integrates industrial robot arms from the major manufacturers into factory production lines, then troubleshoots them to keep operations running.

**Perception Engineer:** Builds the computer vision and sensor systems that let robots recognize objects, map their surroundings, and navigate using cameras and lidar.

**Robotics Controls Engineer:** Designs and tunes the control algorithms and motion-planning routines that make a robot move smoothly, accurately, and safely through its tasks.



## Skillset Requirements

More than in most fields, a robotics engineer has to be a generalist. Strong programming ability matters most, because a robot is only as capable as the software running it, but engineers also need the hands-on skill to wire a sensor, read a circuit, and coax uncooperative hardware into working. Their signature strength is debugging; when a robot misbehaves, the fault may be mechanical, electrical, or buried in the code, and a good engineer hunts it down calmly across all three. A careful eye for safety, creativity for open-ended problems, and the ability to work closely with other engineers round out the set.



## Career Trajectory

New robotics engineers usually start on a focused piece of a larger system, such as integrating a single sensor, tuning one control loop, or writing test scripts for a robot's behavior. As they gain experience, they take ownership of whole subsystems like perception or navigation and eventually architect complete robots from concept to deployment. Some specialize in a domain such as surgical, warehouse, or field robotics, while others move toward research roles pushing autonomy forward, or into technical leadership guiding multidisciplinary teams.



## Pace of Innovation

Robotics moves quickly, pushed forward by steady gains in computing and artificial intelligence. Better and cheaper sensors, faster onboard processors, and smarter algorithms keep expanding what robots can sense, and do, letting them work in cluttered, unpredictable spaces that once defeated them. Learning-based methods are changing how robots perceive and make decisions, so that machines can adapt rather than follow fixed scripts. Because the tools evolve fast, engineers in this field expect to keep learning throughout their careers.



# The Life of a Robotics Engineer

**The Work:** Robotics engineers design, build, and program robots, machines that combine moving parts, electronics, and software, to carry out physical tasks on their own. The result is a machine that can sense its surroundings and act on them safely and accurately.

## Applications

Robotics reaches into almost every industry. On factory floors, robot arms weld, paint, and assemble products faster and more consistently than people can. In warehouses, fleets of mobile robots pick and move goods around the clock. Hospitals use surgical robots that let doctors operate with extraordinary precision, and farms deploy robots to plant, weed, and harvest crops. Other robots inspect pipelines, defuse explosives, and explore places too dangerous for people—from the deep ocean to Mars. Self-driving cars, delivery drones, and the vacuum in many homes are robots too. Wherever a job is dull, dirty, or dangerous, someone is building a robot to do it.

## Knowledge Base

To build a working robot, an engineer has to understand several fields in depth. Mechanics comes first; kinematics and dynamics describe how a robot's joints move, how far an arm can reach, and how much force a motor must supply. Control theory keeps that motion smooth and accurate, using feedback loops that correct small errors hundreds of times a second. Engineers also need electronics, to link motors, sensors, and microcontrollers, and a solid software foundation in algorithms and real-time programming. Artificial intelligence increasingly completes the picture, giving robots the machine learning and computer vision they use to recognize objects and navigate a changing world.

## Interdisciplinary Overlap

Robotics work brings several subdisciplines together. Robotics engineers collaborate with mechanical engineers who design the chassis and moving parts, and with electrical engineers who handle power, wiring, and circuit boards. They partner with software and AI specialists to build perception and decision-making systems, and work with controls engineers to fine-tune motion. On the factory floor, they coordinate with manufacturing and industrial engineers to fit robots into production lines. Because their machines touch so many fields at once, robotics engineers spend much of their time coordinating across teams and translating between the mechanical, electrical, and software sides of a project.

## Work Environment & Lifestyle

A robotics engineer's time is split between the computer and the lab. Much of the day goes to writing and debugging code, running simulations, and reviewing designs. But the work is hands-on, so engineers also spend many hours in the lab building prototypes, wiring sensors, and testing robots on the floor. Some travel to customer sites or factories to install and commission systems.

Hours usually follow regular weekdays, though a looming demo or a tricky bug can stretch into long evenings.

The work is demanding, but watching a robot you built finally work is deeply rewarding.

## A Day in the Life

### Meet Mateo, a warehouse robotics engineer

Mateo is a robotics engineer at a warehouse automation company, where his days move between writing code at his desk and testing the robots out on the warehouse floor.

**8:00 AM:** Mateo starts his day reviewing logs from an overnight test run. One of the picking robots stopped mid-shift and the error trail points to its camera losing track of items in dim light. He pulls the robot into the lab, reproduces the failure, and traces it to a vision setting tuned too aggressively. He adjusts the parameters, reruns the test, and watches the robot identify each item reliably before returning it to service.

**10:30 AM:** Mateo joins a design review for a new robotic arm meant to handle heavier packages. Using simulation software, he checks whether the chosen motors can move the arm quickly without overshooting its target. The model shows a slight wobble at full speed, so he proposes tuning the control gains and adding a short damping step. The team agrees to build a prototype and put his fix to the test.

**1:30 PM:** After lunch, Mateo pairs with an electrical engineer to track down a sensor that keeps dropping out on a test robot. They find that a loose connector and electrical noise from a nearby motor are corrupting the signal. Mateo reroutes the wiring and adds a software filter, and the readings finally hold steady through a full test cycle.

**4:30 PM:** Back at his desk, Mateo writes automated tests so the morning's vision bug can't slip through unnoticed again. He documents the fix, pushes his code to the shared repository, and reviews a teammate's work before wrapping up. With the picking robots running smoothly again he closes his laptop for the day.

Mateo heads home knowing that a fleet of robots will keep moving through the night because of the fixes he made today. The robots make the work look easy, but he knows that kind of reliability is earned one careful test and one stubborn bug at a time.

# DATA & AI ENGINEERING

*"If a computer recognizes your voice or predicts what you'll like, a data and AI engineer trained it to learn."*

**Data and AI engineers develop the intelligent systems** that allow computers to learn from data. They blend software engineering with mathematics and domain knowledge to create algorithms and tools that process raw information and transform it into actions. From recommendation engines to voice assistants, these engineers bridge cutting-edge machine learning research with real-world applications, ensuring that AI technologies are reliable, scalable, and solve practical problems. Ultimately, they play a key role in bringing AI from the lab into everyday life.

## Fast Facts

**Education:** Electives in machine learning, algorithms, and cloud computing; master's degree is beneficial.

**Key Skills:** Strong foundations in statistics and linear algebra, proficient in programming languages, solid grasp of machine learning algorithms and data science, systems thinking, and creative problem-solving.

**Typical Employers:** Major tech companies, R&D units in healthcare, finance, energy, research labs, universities, AI startups, and consulting firms.

**Personality Fit:** Analytical, curious, and experimental, persistent, comfortable with iterative development and exploring new ideas.

**Salary Range:**

$ $ $ $ $ $ $ $ $ $

**Work Environment:**

**Office**

## Job Options

**Machine Learning Engineer:** Designs algorithms that learn from data, building predictive models and integrating them into real-world software applications.

**AI Systems Engineer:** Integrates AI components into broader platforms, ensuring that models operate smoothly within complex systems and scale effectively.

**Data Engineer:** Develops and maintains the infrastructure that supplies clean, organized data to AI systems, optimizing pipelines for speed, reliability, and scalability.

**Data Scientist:** Applies statistics and machine learning to analyze data, uncover trends, and generate insights that inform decisions and drive product features.



## Skillset Requirements

A data and AI engineer needs a blend of advanced math, coding expertise, and system design know-how. They use statistical modeling and probability theory to make sense of data and shape how algorithms learn. Solid software engineering abilities are essential for writing efficient code and handling large datasets. Systems thinking helps them design end-to-end pipelines, from data ingestion and processing to model training and deployment. Equally important is an experimental mindset as they constantly iterate and tune models, trying different approaches that improve results. Communication skills also come into play as these engineers often need to explain complex findings or limitations to non-experts.



## Career Trajectory

Many data and AI engineers begin by developing machine learning research, or in data science roles, where they delve into developing new algorithms. Some become AI infrastructure architects or MLOps leads, building platforms that let many models run reliably at scale. Others go domain-specific, applying AI expertise to a particular industry, like healthcare or finance. As engineers gain experience, many move into leadership positions, or advanced research roles, guiding teams and pushing the state-of-the-art techniques.



## Pace of Innovation

Data and AI engineering evolves at breakneck speed—what was cutting-edge one year might become outdated the next. Engineers must be lifelong learners to keep up with constant advances and new tools. The fast pace is exciting but demanding, nearly every week there's a new library or model architecture to learn. For those drawn to this subdiscipline the constant change is exhilarating, but it also means staying curious and adaptable to keep up.



# The Life of a Data & AI Engineer

**The Work:** Data and AI engineers turn raw data into intelligent systems. They build the pipelines and algorithms that let computers recognize patterns, make predictions, and improve—essentially teaching machines to perform tasks that normally require human intelligence.

## Applications

These engineers apply AI across many real-world projects. They might build a model to predict equipment failures in a power grid or train a computer vision system to detect diseases from medical scans. On another project a data and AI engineer could be crafting a recommendation algorithm for a streaming service or refining the perception system of a self-driving car. Beyond creating the model, they handle the practical steps too: preparing and cleaning data, choosing appropriate algorithms, and training AI models for deployment into software or devices. The goal is AI solutions that aren't just clever, but also reliable and useful in daily life.

## Knowledge Base

Data and AI engineers draw on a broad foundation of math and computer science. They need a strong grasp of statistics and probability, as well as plus linear algebra and calculus, to understand data and how algorithms work. They must also be skilled coders proficient in multiple programming languages, well-versed in data structures, and comfortable using cloud computing to manage large datasets. Experience with machine learning libraries and frameworks helps them build and test models efficiently. Finally, they must know the domain they're working in, whether it's healthcare, energy, or another field.

## Interdisciplinary Overlap

Data and AI engineers rarely work alone. They collaborate with other data scientists and researchers to turn prototype algorithms into scalable solutions, and partner with software developers and IT teams to integrate AI models into broader systems or cloud services. They also coordinate with data engineers to ensure the AI systems have clean, well-structured data, and work with domain experts or product managers to make sure the technology actually solves the intended problem. In short, their role serves as the bridge between cutting-edge ideas and practical deployment, requiring them to work collaboratively across multiple teams on every project.

## Work Environment & Lifestyle

Data and AI engineers typically work in offices or tech labs, spending much of the day coding. Collaboration is constant, with teams brainstorming solutions together, reviewing code, and discussing results from experiments together. The workflow is highly iterative: they code, test, analyze results, and refine their approach. The pace is brisk, with tight project deadlines and rapidly evolving technology, but the environment rewards creativity and problem-solving. Hours are usually regular weekdays, though occasionally a deadline or long training run requires extra hours.





## A Day in the Life

### Meet Walter, a data and AI engineer

Walter is a data and AI engineer at a renewable energy company where he builds intelligent systems that optimize how wind turbines and solar panels feed power into the electric grid.

**9:00 AM:** Walter begins his day with a quick team stand-up to share updates. After the meeting, he dives into debugging a failed overnight data pipeline run. He finds a format error in the latest sensor data file and fixes the code. Once he re-runs the pipeline, the data flows properly again, getting the energy output prediction model back on track by mid-morning.

**11:00 AM:** Later, Walter meets with an energy analyst and the product manager to check how an AI model is performing. They notice the model's solar output predictions were inaccurate when cloud cover rolled in suddenly the previous day. The team decides to feed real-time weather data into the model to improve responsiveness.

**2:00 PM:** After lunch, Walter updates the system to include the weather feed and kicks off a fresh training run on the model. While the model trains on the expanded dataset, he double-checks that the incoming weather data is flowing correctly. By late afternoon, training is complete and his validation tests show the updated model is much more accurate, especially during sudden weather changes.

**4:00 PM:** Pleased with the improvement, Walter deploys the updated model to a staging environment, and he writes a brief note to the energy analyst and product manager about the bug fix and the new weather feature. He schedules a follow-up call with the weather data provider to learn more about their data sourcing and accuracy.

Walter leaves knowing that his work directly improves the reliability of renewable energy systems. By pushing sensor readings into data sets, he helps create the automated systems that make our world smarter.

# Starting the Conversation

A Discussion Guide for Parents and Counselors



## How to use this section

**You don't need to know a thing about engineering** to use what follows. That's the whole idea. This isn't a test, there's nothing to score, and there are no right answers to add up. It's a set of conversation starters—questions you ask your student, each with a short note telling you what a promising answer sounds like, so you'll recognize it when you hear it.

**The one worry to clear out of the way first.**
Most people assume engineering is a club for kids who are naturals at math, and that if your student isn't at the top of the math class the door is shut. It isn't. Math is a skill that gets taught, the same way reading is, and plenty of strong engineers were ordinary math students who grew into it. What marks a future engineer is something else: wanting to know why things work, sticking with a hard problem instead of bailing on it, and looking at things and itching to build them better. If your student has those, the math will come. Watch for those, not for grades.

Two questions, not one. You're really helping with two different decisions, and it's worth keeping them apart. The first is whether engineering fits this kid at all—that's Stage One. The second, worth asking only when the first looks promising, is which kind of engineering—that's Stage Two, with a short section for each of the five families.

How to do this without steering. The goal is to get your student talking and thinking, not to land them on an answer by the end of the afternoon. The most useful thing you can do is ask, then get quiet and let them work it out loud. If you catch yourself steering toward the answer you're hoping for, ease off, a path they reason their way into will hold up, and one you talk them into won't.



**If you're a counselor:** you already run conversations like this every day; what you've been missing is what to listen for in this one specific field, and that's what the notes give you.

**If you're a parent:** your advantage is that you've watched this kid their whole life. Use it. The best openers below are the ones where you can say "remember when you…"—start there.

**A note on the parent prompts:** the specific memories in them (the toaster, the creek, the crane downtown) are stand-ins. Swap in your own—the real memory from your kid's life will always work better than the example on the page.

# STAGE ONE: Is there an engineer in there?

Eleven things to listen for. Don't treat any single answer as a verdict—you're reading a pattern across all of them. If several of these light your student up, Stage Two is worth doing together.

## 1 Wanting to know why

**Counselor** Think of the last thing that broke or quit on you; your car, your phone, something in your house. Were you curious about why it failed?

**Parent** Remember when you took apart the toaster? Take me back—were you trying to fix it, or did you just want to see what was inside?

**Listen for** They pull toward the cause, not just the cure. A kid who wants to understand the mechanism, even when fixing it, is showing you the most basic engineering reflex there is.

## 2 Staying with a hard problem

**Counselor** Tell me about something hard you stuck with longer than you probably should have. What kept you in it?

**Parent** I remember you spending a whole weekend stuck on one thing. What was going through your head while you couldn't crack it?